Research on performance verification, evaluation and analysis of multi-parameter cardiac monitor

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Abstract

With the support of market demand and relevant national policies, the scientific instrument industry has shown a rapid development trend, the overall strength of China's scientific instruments has also been greatly improved, and a number of domestic scientific instruments with independent intellectual property rights are rising. However, the progress of technology hasn't brought about the growth of sales of domestic scientific instruments, and the market recognition is still relatively low. Some scientific research and institutions don't understand and daren't buy them. This study explores verification and evaluation methods to find the advantages, improve the popularity and market recognition and achieve comprehensive promotion of multi-parameter cardiac monitor. Taking multi-parameter cardiac monitor as the representative, we have developed the comprehensive verification and evaluation scheme of the instrument, carried out the performance comparison of instruments at home and abroad, formed the verification and evaluation report of domestic scientific instruments through a large number of performance comparison tests and data analysis.

CCS Concepts

• **Computing methodologies** → Modeling and simulation; Simulation evaluation.

Keywords

Verification, Evaluation, Medical metrological instrument

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1 Introduction

In October 2016, the CPC Central Committee and the State Council jointly released an outline for the "Healthy China 2030" initiative, specifying the targets and programs of action. The Party decided to carry out the "Healthy China initiative" at its 19th National Congress, a move to lay a solid foundation for building a prosperous society and turning China into a great modern socialist country that is prosperous, strong, democratic, culturally advanced, harmonious and beautiful.

In the context of the healthy China 2030 strategy, the global precision medical plan is in full swing. In the face of the intensification of population aging and the innovation of medical technology, people's demand for health and medical services is increasing day by day, so as to protect people's health and implement the disease prevention strategy Improving medical and health services and improving the technical capacity and level of household medical monitoring equipment have become an important technical support to meet people's aspirations for a healthy China.

Multi-parameter cardiac monitor is a kind of common medical equipment that can detect ECG, heart rate, oxygen saturation, blood pressure, respiratory rate, body temperature and other important parameters of human body in real time through various functional modules, so as to realize the supervision and alarm of various parameters [1-3]. As the main scientific instrument supporting "accurate diagnosis" and "accurate medication" in the construction of precision medical system, multi-parameter cardiac monitor plays a vital role in monitoring the parameters of human physiological signs.

On the one hand, the work of Carrying out the performance verification and evaluation of domestic multi-parameter cardiac monitors, releasing the comprehensive evaluation results of the performance of domestic instruments and imported instruments, establishing the measurement and detection capability represented by multi-parameter cardiac monitors, looking for the advantages and disadvantages of domestic instruments, and forming a verification and evaluation service platform through technical improvement and product upgrading, could promote the product quality of domestic instruments and improve product competitiveness [4-8]. On the other hand, it can make a scientific and fair comprehensive performance evaluation on the performance of domestic equipment and imported equipment, let all sectors of society see the

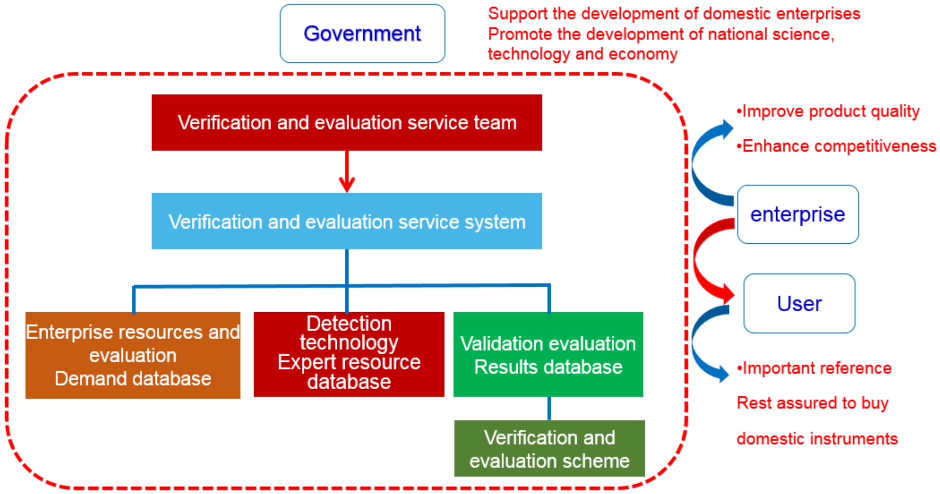

Figure 1: verification and evaluation service platform.

Figure 2: multi-parameter cardiac monitor.

progress of domestic equipment technology, form an effective product boost [9-10], and improve the market share of domestic instruments, which has great economic and social significance (Figure 1).

2 Multi-parameter cardiac monitor

The multi-parameter cardiac monitor consists of three parts: signal acquisition, signal processing and signal display output. The signal acquisition part includes a sensor and an ECG electrode to convert the biological signal into a measurable electrical signal; The signal processing part includes analog processing and digital processing; The signal display and output part includes digital display, waveform display, copy recording, audible and visual alarm, etc.

Clinically, the multi-parameter cardiac monitor can provide comprehensive data information. It has the characteristics of real-time data monitoring, timely alarm, convenient playback and record management (Figure 2). It is more and more favored by medical institutions, and the number of use is increasing year by year. Its main use significance lies in: first, critically ill patients must monitor them, report to the police immediately and notify doctors to rescue them in time; Second, the occurrence time of some disease phenomena is short, and it takes a long time to record the abnormal phenomena.

With the rampant New Coronavirus, the number of confirmed patients and suspected patients is increasing. The multi-parameter cardiac monitor plays an irreplaceable role in the prevention and control of epidemic situation. "WHO 2019 New Coronavirus guidelines" points out that 2019-nCOV epidemic can be shown as mild, moderate and severe. The latter includes severe pneumonia, hypoxic respiratory failure and acute respiratory distress syndrome (ARDS), sepsis and septic shock. Blood oxygenation is one of the

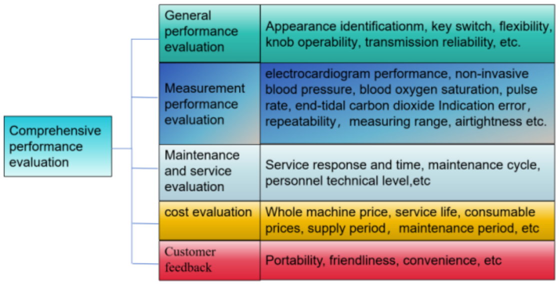

Figure 3: Verification and evaluation project content.

diagnostic criteria for ARDS. Therefore, the wide use of multi-parameter cardiac monitor provides an accurate and efficient technical guarantee for the diagnosis and course monitoring of clinical severe patients.

In the face of such a large number of medical products closely related to human life, the importance of their quality is self-evident. How to improve the core quality of domestic equipment, boost people's confidence in domestic products, and improve the international market competitiveness of products is the goal and direction of this topic [11-13].

For a long time in the past, high-end medical devices and equipment have been monopolized by a few foreign enterprises. Foreign products still occupy the main market of China's tertiary hospitals. Some core components of domestic brands can't be produced in China and need to be imported. However, in recent years, domestic high-end medical devices and equipment have sprung up [14-16].

3 Analysis of test scheme and evaluation

3.1 Performance evaluation content

At present, although the verification regulation of multi-function patient monitoring instruments (JJG 1163-2019) has been promulgated and implemented, the measurement work of multi-parameter cardiac monitors has not been fully carried out, which has partially led to a lack of supervision of the quality of multi-parameter monitors and the inability to achieve traceability and uniformity of measurement values.

In order to promote the technological progress of multi-parameter cardiac monitors and eliminate dependence on imported measuring instruments, this article conducts scientific and fair quality verification and evaluation of multi-parameter cardiac monitors [17]. By identifying technological gaps, improving products, and enhancing the quality of domestic products. The evaluation content of this article mainly includes the following five aspects, namely general functions, measurement performance, customer experience, maintenance and service evaluation, and cost evaluation. Through joint testing by multiple testing technology institutions, we strive to comprehensively and objectively evaluate the instrument (Figure 3).

The measurement performance evaluation of multi-parameter cardiac monitors can refer to the verification regulation of multi-function patient monitoring instruments, including electrocardiogram performance, non-invasive blood pressure, blood oxygen saturation, pulse rate, end-tidal carbon dioxide, etc. In addition to metrological performance, it is also necessary to assess the appearance, safety performance, ease of operation, instrument maintenance, after-sales service, service life, and purchase cost of the instrument.

3.2 Result analysis and research

The instruments participating in the verification and evaluation meet the requirements in terms of appearance, all instrument appearance markings, all fasteners are installed firmly, connected well, and there is no looseness. The buttons and switches can work normally without obvious vibration or sound, and have real-time waveform display function and strong scalability.

The key devices of a multi-parameter cardiac monitor are the electrocardiogram module, blood pressure stimulation, and blood oxygen system. Different brands of instruments have differences in blood pressure module or blood oxygen curve system. The principle of a multi -parameter cardiac monitor is to pick up physiological parameter signals such as human electrocardiogram, blood pressure, blood oxygen, end of breath, and body temperature through electrodes and sensors, and convert these signals into electrocardiogram signals (Figure 4). The monitor consists of four components: signal parameters, analog processing, digital processing, and information output [18-19].

Before the experiment, the indoor quality control test results of the instruments were reliable and had good repeatability. In this experiment, the two instruments did not have any faults, had good stability, was easy to operate, tested quickly, and the measurement performance index error met the quality control requirements of the Ministry of Health for multi-parameter monitoring testing.

Before the experiment, the indoor quality control test results of the two instruments were reliable and had good repeatability. In this experiment, there were no malfunctions in the two instruments, and they had good stability, easy operation, and fast detection.

The two instruments, using the same standard and testing personnel, showed good repeatability of the measurement performance indicators of the two multi-parameter cardiac monitors, and the measurement performance indicator error met the quality control requirements of the Ministry of Health for multi-parameter monitoring and testing.

In terms of heart rate indication error, static pressure measurement range, static pressure indication error, blood pressure indication repeatability, static pressure indication stability, pulse oxygen saturation indication repeatability, pulse rate indication error, end tidal carbon dioxide concentration indication repeatability, and respiratory rate indication error, the comparison results between Beijing M&B (model MB526T7) multi-parameter cardiac monitor and PHILIPS (model 866062) model show that the measurement performance indicators of the two multi-parameter monitors are higher than the standard indicators, both within the required range, with good linearity. Both instruments have good accuracy, and the measurement results are accurate and reliable (Table 1). The comparison of the measured values between the two instruments,

<?TeX R = 0.98 ?>

Figure 4: Verification and evaluation project content.

Table 1: Evaluation results for measurement performance indicators.

Manufacturer			Beijing M&B	Germany HILIPS
model			MB526T17	866062
ECG	Voltage measurement error		0%	0%
	Scanning speed error		0%	0%
	Amplitude frequency characteristic		0%	0%
	Heart rate error		-1 BPM	0 BPM
Non-invasive blood pressure	Static pressure measurement range		(0~260) mmHg	(0~260) mmHg
	Static pressure error		-1 mmHg	0.6 mmHg
	Repeatability	Systolic pressure	0.4 mmHg	0.9 mmHg
		Dilational pressure	0.9 mmHg	1.7 mmHg
	Stability		0 mmHg	0.9 mmHg
	Airtightness		0.4 mmHg/ min	1.0 mmHg/ min
Pulse Oximetry	Repeatability		0 %	-1 %
	Pulse rate error		0 BPM	0 BPM
$EtCO_2$	Repeatability of concentration indication		-0.1 mmHg	-2.2 mmHg
	Respiratory rate error		-1BPM	0BPM

R value indicates that the two instruments have a good linear correlation and regression relationship, and the results are comparable. The above performance verification experiments demonstrate that both instruments meet the quality objectives of the laboratory and meet the requirements of clinical and scientific research for performance testing.

In order to comprehensively analyze the performance level of the multi-parameter cardiac monitor, this article conducts a comprehensive evaluation and analysis of the measurement performance based on the measurement data obtained from the test results, uses questionnaire surveys and customer experience analysis based on user experience, and evaluates and compares the general functional characteristics, maintenance and service evaluation, and consumables cost. The comparative analysis radar evaluation chart shown in Figure 5 is obtained.

In terms of indication error, domestic multi-parameter cardiac monitors need to continuously innovate in technology, including improving the accuracy of non-invasive blood pressure dynamic measurement, airtightness, and the diversity of blood oxygen saturation curve [20]. By adapting to the physiological sign information characteristics of the Chinese population, we can improve the technical adaptability from multiple aspects and levels, and better improve relevant overall metrological performance indicators of domestic monitors.

4 Conclusion

To sum up, multi-parameter monitors and other biomedical instruments can promote technological progress and innovative development through continuous comparative analysis, technological improvement, and validation evaluation, narrow the gap between manufacturing enterprises and manufacturers in the field of instruments, and continuously improve the technical level of biomedical instruments [21-22]. At the same time, we should increase investment in scientific research and innovation, timely understand and grasp the needs of users, transform the needs into the driving force for product research and development, completely eliminate the dependence of the domestic market on imported measuring instruments, improve the quality competitiveness and reliability of domestic quality supervision and measurement scientific instruments, strengthen government or technical institution procurement intentions, let domestic and foreign users understand the quality

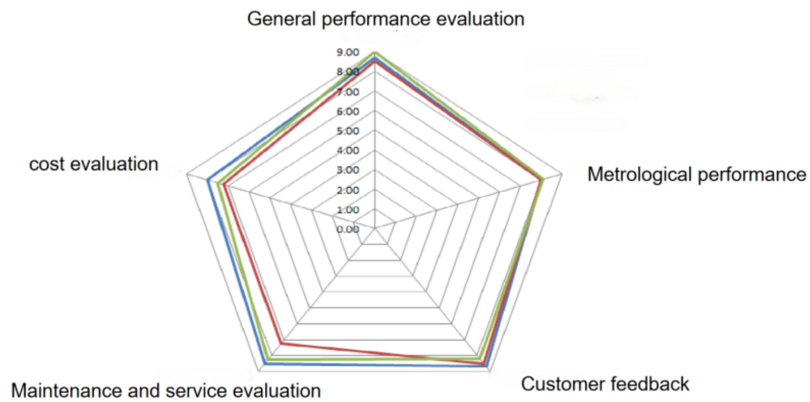

Figure 5: Comparative analysis of radar charts.

and technology of domestic scientific instruments, and achieve the grand goal of "Quality China World Tour".

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